

Power BI Project - Transforming EDAs to Dashboards

1. Project Title

Amazon Prime Video Content Strategy & Analysis.

2. Dataset Name and Description

Dataset Name

Amazon Prime Video Movies and TV Shows Dataset

Dataset Description

The dataset available as two csv file **TITLES** and **CREDITS**. Basically, **titles** contains the content (TV Shows or Movies) related details such as title, type, genres etc. Whereas **credits** contains the actors/directors details that have got a role on each title. The id (title id) field is the common key field between these 2 sets that can be used to join/merge these 2 dataframes.

Dataset Description

Dataset 1: titles_final

This dataset contains detailed information about movies and TV shows available on the streaming platform.

Column Name	Data Type	Description
id	Integer / String	Unique title identifier on JustWatch (Primary Key).
title	Text	Name of the movie or TV show.
show_type	Text	Type of content (Movie or TV Show).
description	Text	Brief synopsis or description of the title.
release_year	Integer	Year in which the title was originally released.
age_certification	Text	Content maturity rating (e.g., PG, PG-13, R, TV-MA).
runtime	Integer	Duration of the movie or episode in minutes.

Column Name	Data Type	Description
genres	List / Text	List of genres associated with the title (e.g., Drama, Horror, Comedy).
production_countries	List / Text	Country codes representing production countries (e.g., US, CA, IN).
seasons	Integer	Number of seasons available (applicable only to TV Shows).
imdb_id	Text	Unique identifier of the title on IMDb.
imdb_score	Decimal	IMDb user rating score.
imdb_votes	Integer	Total number of votes received on IMDb.
tmdb_popularity	Decimal	Popularity score calculated by TMDb.
tmdb_score	Decimal	Average user rating score on TMDb.

Primary Key: id

Dataset 2: credits_final

This dataset contains information about actors and directors associated with each title.

Column Name	Data Type	Description
person_ID	Integer / String	Unique identifier of the actor or director on JustWatch.
Id	Integer / String	Title identifier used to link with the df_titles dataset (Foreign Key).
Name	Text	Name of the actor or director.
character_name	Text	Character portrayed by the actor in the title.
Role	Text	Role of the individual in the title. Values include ACTOR or DIRECTOR.

Foreign Key: id → References titles.id

Data Model Relationship

Parent Table **Child Table** **Join Key** **Relationship Type**

Titles	credits	id	Many-to-Many(M:M)
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Relationship Explanation

- One title can have multiple actors and directors.

- Each actor/director record can belong to one or more titles.
- The datasets are joined using the **id** column.

3. Dashboard Objective

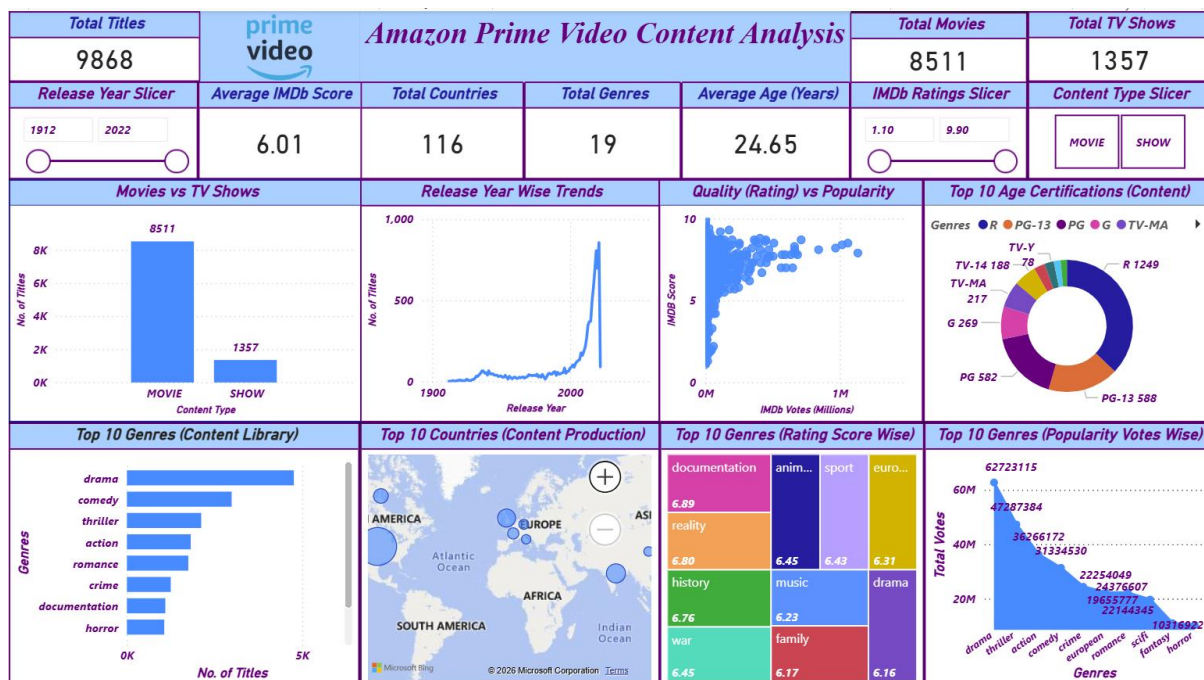
The objective of this project is to provide Amazon Prime Video stakeholders with data-driven insights to support content acquisition, portfolio optimization, audience coverage, and global expansion strategies.

The dashboard helps answer the following business questions:

- Which genres dominate the content catalog?
- Which countries contribute the most content?
- Are there content gaps across regions and genres?
- How fresh is the current content portfolio?
- What opportunities exist for future content acquisition?
- How can Amazon improve audience engagement through a balanced content strategy?

4. Dashboard Screens

Screen 1: Executive Overview



Purpose:

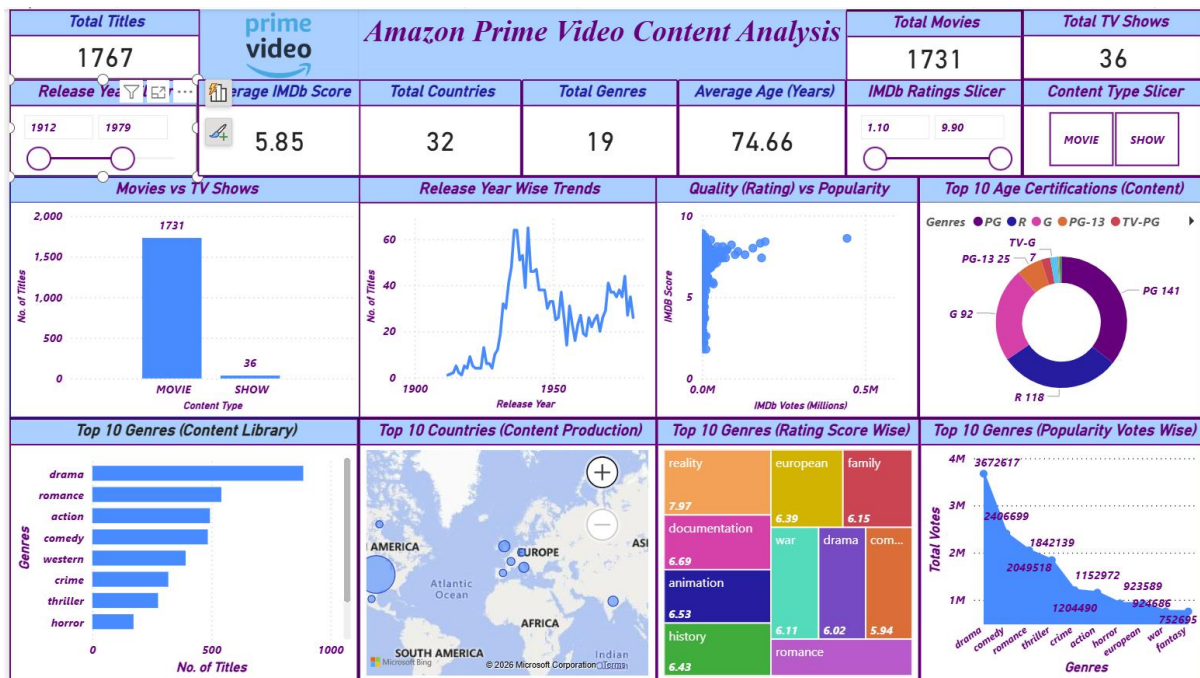
Provides a high-level overview of the content catalog including total titles, movies, TV shows, genres, countries, and growth trends.

Key Visuals:

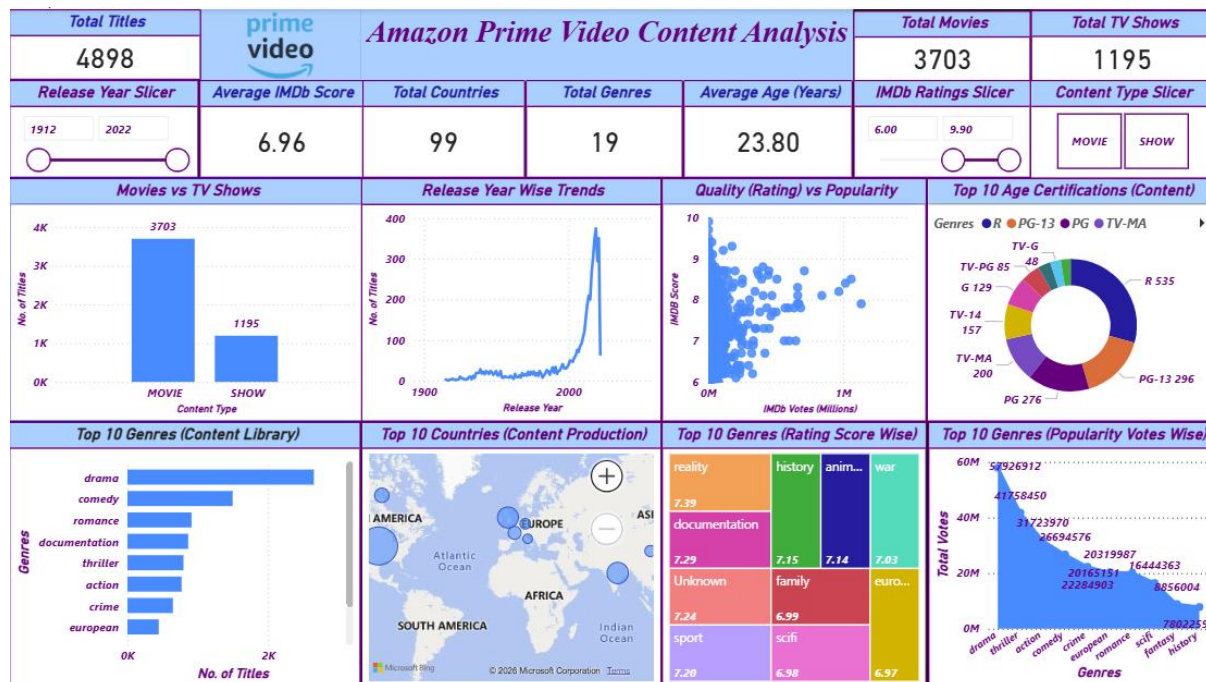
- KPI Cards
 - Content Growth Trend
 - Movies vs TV Shows Distribution
 - Top Genres
 - Top Countries
 - Slicers
-
- **KPI cards** are used because they provide a **quick summary** of the platform at a glance. Decision-makers can quickly understand the size, quality, diversity, and freshness of the content library without interpreting charts.
 - **Slicers** enable users to interactively analyze how the content library changes over different parameters like release year etc.
 - A **scatter plot** is used to analyze the relationship between two numerical variables—IMDb Rating and IMDb Votes. It reveals a positive relationship between audience popularity and content quality but at the same time we can also observe significant portion of the catalog consists of low-visibility titles with limited audience interaction
 - A **donut chart** effectively shows the proportion of each age certification within the catalog. It helps understand the target audience distribution (adult vs family-friendly content).
 - A **map** visual is the most intuitive way to display geographic distribution. It helps stakeholders quickly identify where content is being produced and detect regional concentration or expansion opportunities.

Other Screenshots after various user interactions through slicers:

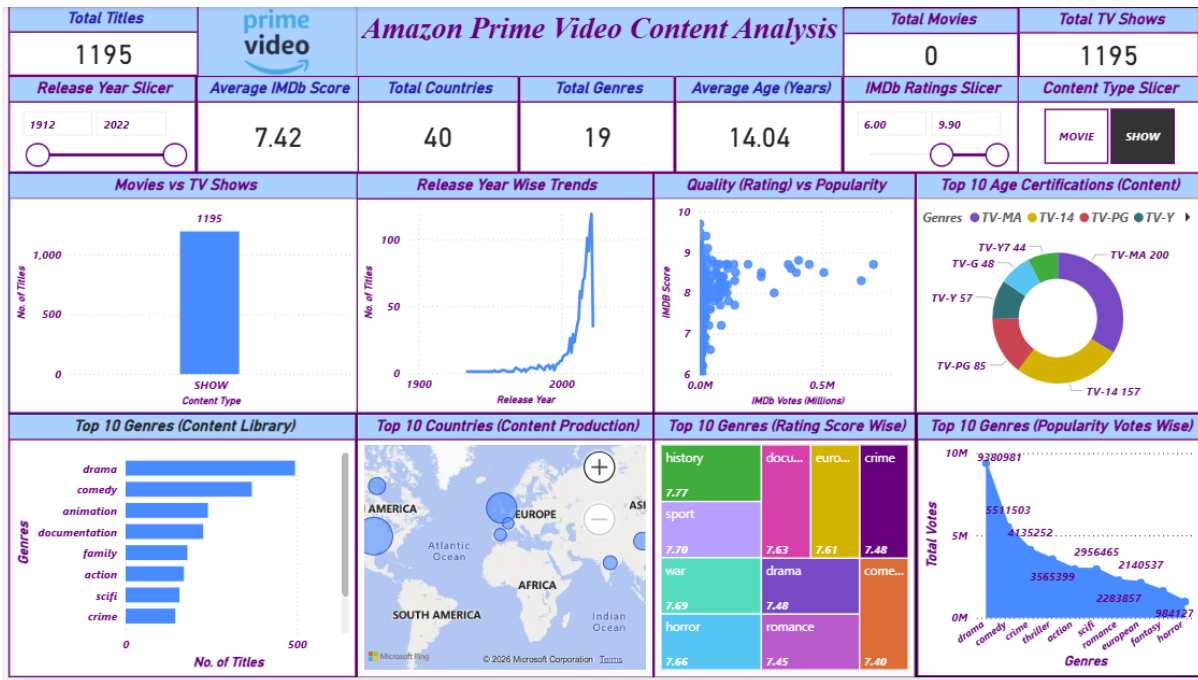
When used slicer to show content with release year ≤ 1979



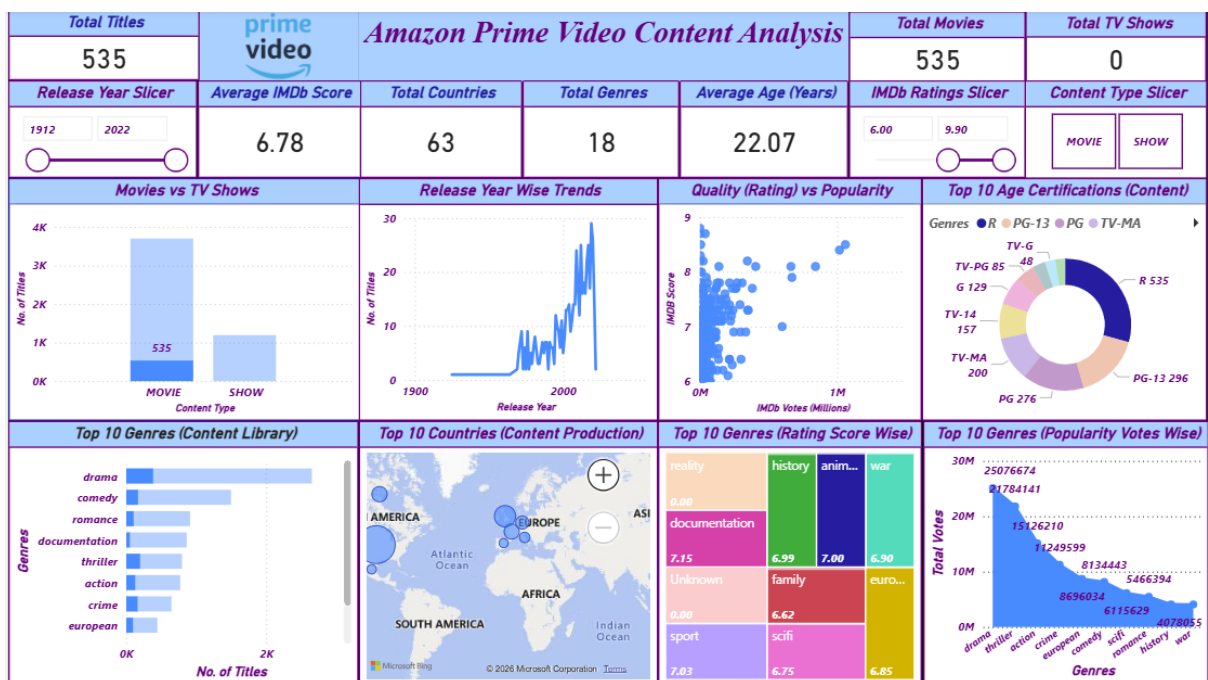
When used slicer to show content with IMDb Ratings Score ≥ 6.0



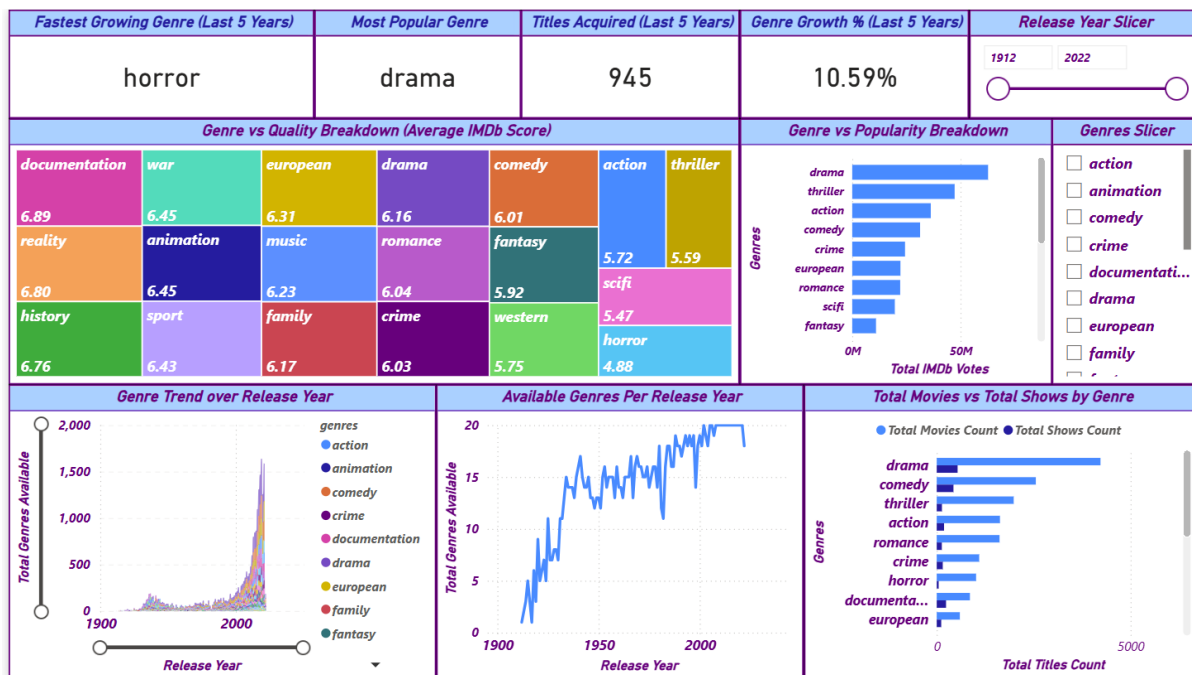
When used slicer to show content with type = TV Shows and IMDb Ratings Score ≥ 6.0



When used slicer to show content with all R rated Content with IMDb Ratings Score ≥ 6.0



Screen 2: Content Acquisition Intelligence



Purpose:

Supports content acquisition decisions by identifying genre and regional content gaps.

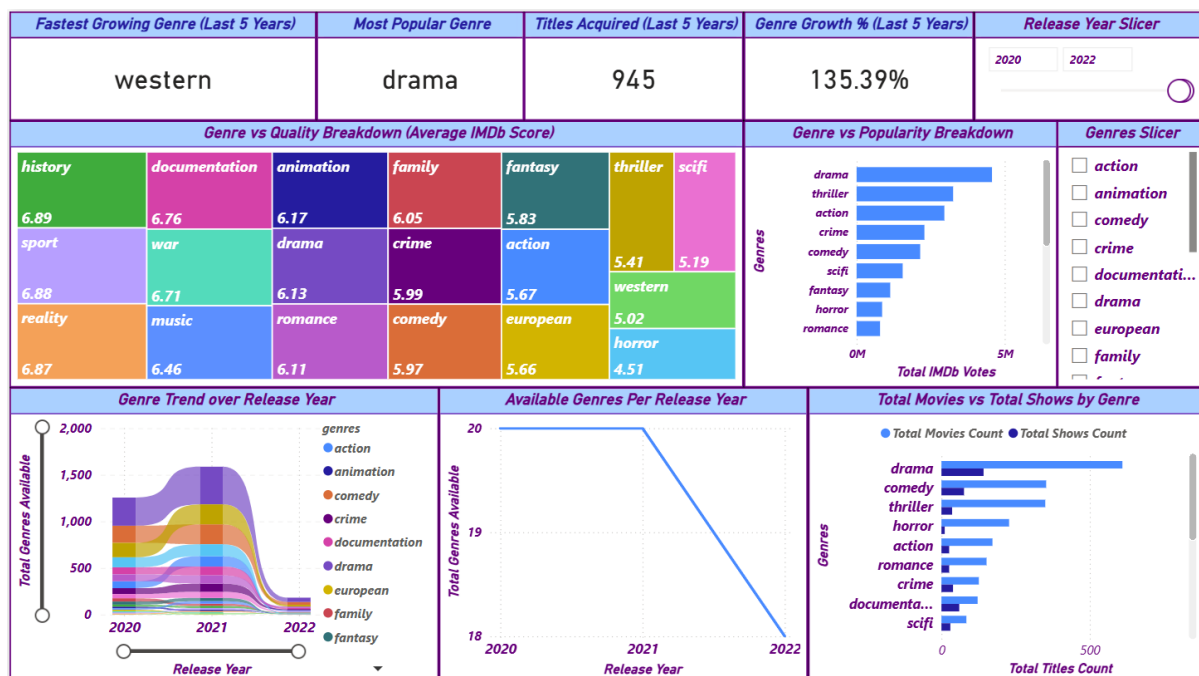
Key Visuals:

- Genre Quality Distribution Treemap
 - Genre Popularity Analysis
 - Genre Trend over Release Year
 - Acquisition Opportunity (Total Movies vs Total Shows by Genre)
- A **treemap** efficiently compares average IMDb ratings across multiple genres while using minimal dashboard space. Larger blocks and labels make it easy to identify high-performing genres at a glance.
 - A **line chart** is the most appropriate visual for showing changes over time. It helps identify whether Amazon Prime is expanding or reducing genre diversity across different years.
 - A **Ribbon chart** shows how genre acquisitions have changed over time while also illustrating each genre's contribution to the total content library. It is effective for identifying growth and decline patterns.
 - **KPI cards** provide an executive overview of genre growth, popularity, and recent acquisitions.
 - **Horizontal bar chart** to compare genre quality and audience engagement because they make ranking and comparison easy.

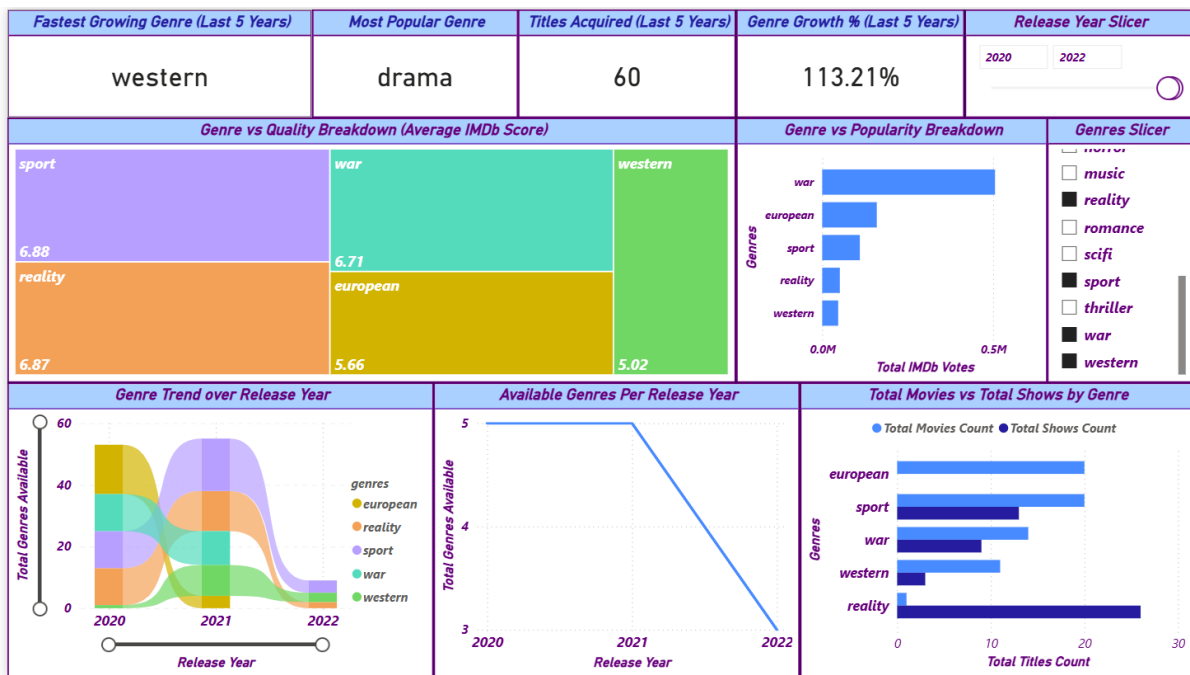
- Finally, a **clustered bar chart** to compare movie and TV show distribution within each genre, helping identify content portfolio gaps.
- Together, these visuals enable decision-makers to understand which genres are growing, which perform best, and where future content investments should be focused.

Other Screenshots after various user interactions through slicers:

When used slicer to show content with release year ≥ 2020



When used slicer to show content with release year ≥ 2020 and genres in (war, reality, sport, western and European)



Screen 3: Global Expansion Strategy



Purpose:

Analyzes geographic distribution of content and identifies expansion opportunities.

Key Visuals:

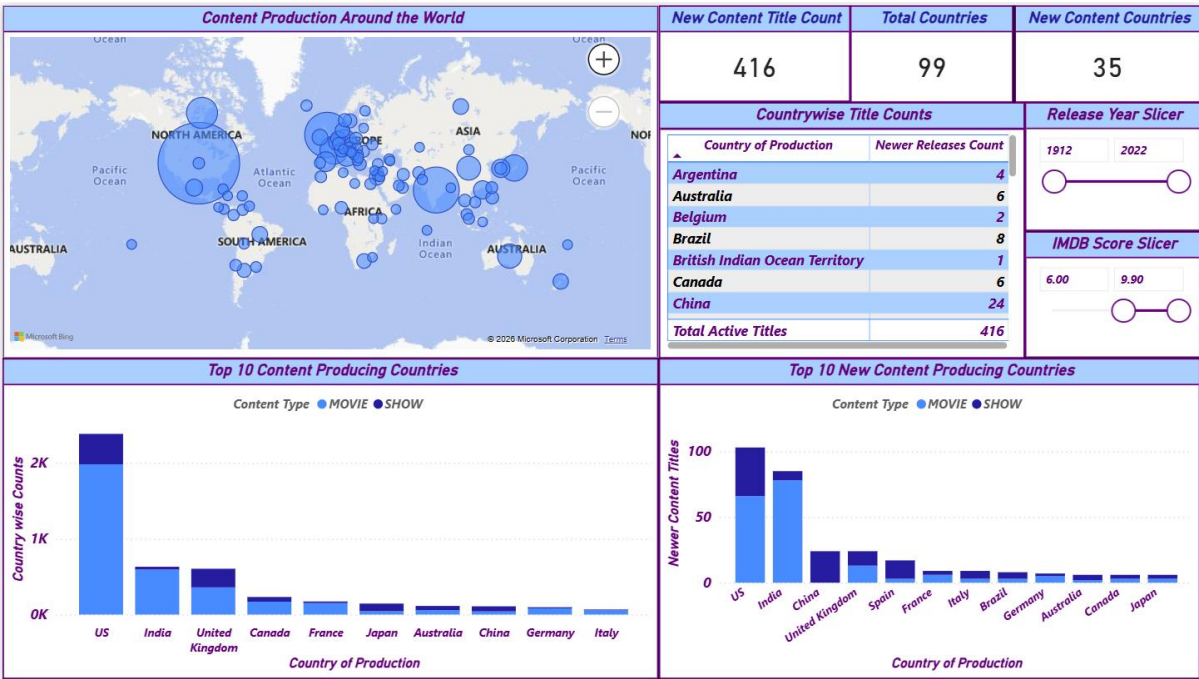
- Global Content Map
 - Country Ranking (Top 10 Content Producing Countries)
 - Regional Distribution Analysis
-
- A **Map** is the most effective visual for displaying geographic data. It allows stakeholders to instantly identify where content is being produced worldwide. Bubble size represents the volume of content produced by each country.
 - A **Table** is used to display country wise recent titles contributed as it allows stakeholders to go deeper into which countries providing recent content. Supports detailed operational analysis.
 - **KPI Cards** useful for quick glance of global outreach. Along with **sliders** it gives business stakeholders meaningful insights
 - **Stacked Column Chart** enables quick TOP N analysis for understanding which countries are leading in content production and in new or recent content production.

Other Screenshots after various user interactions through slicers:

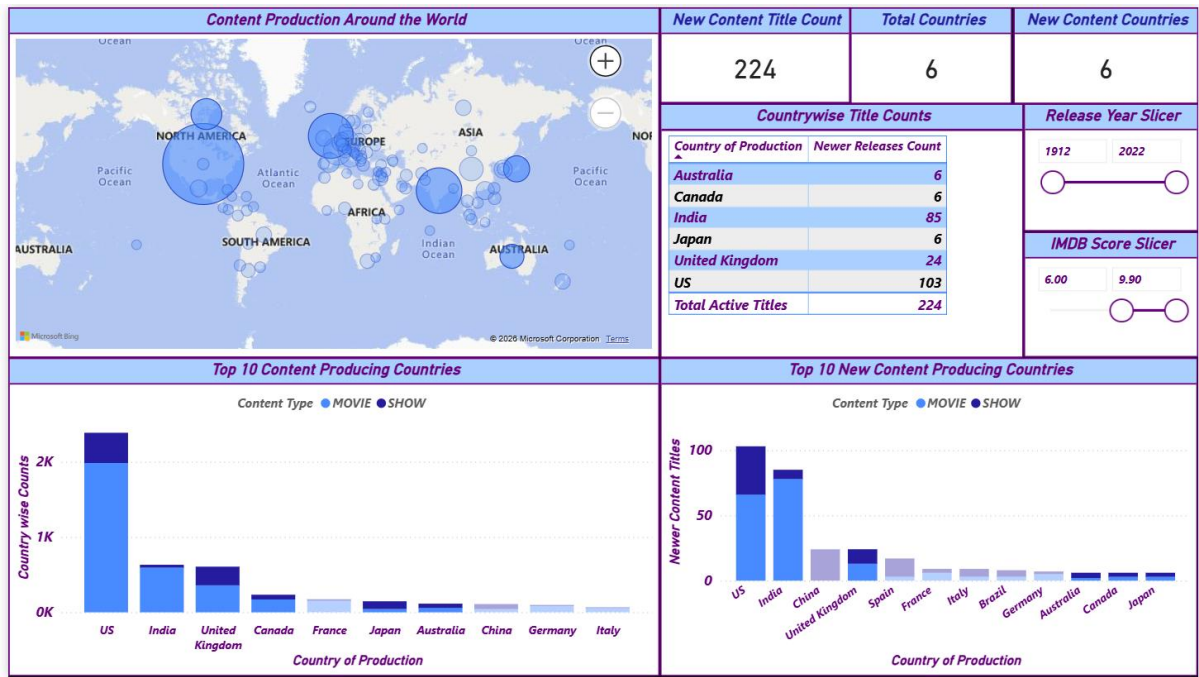
When used slicer to show content with release year ≥ 2020



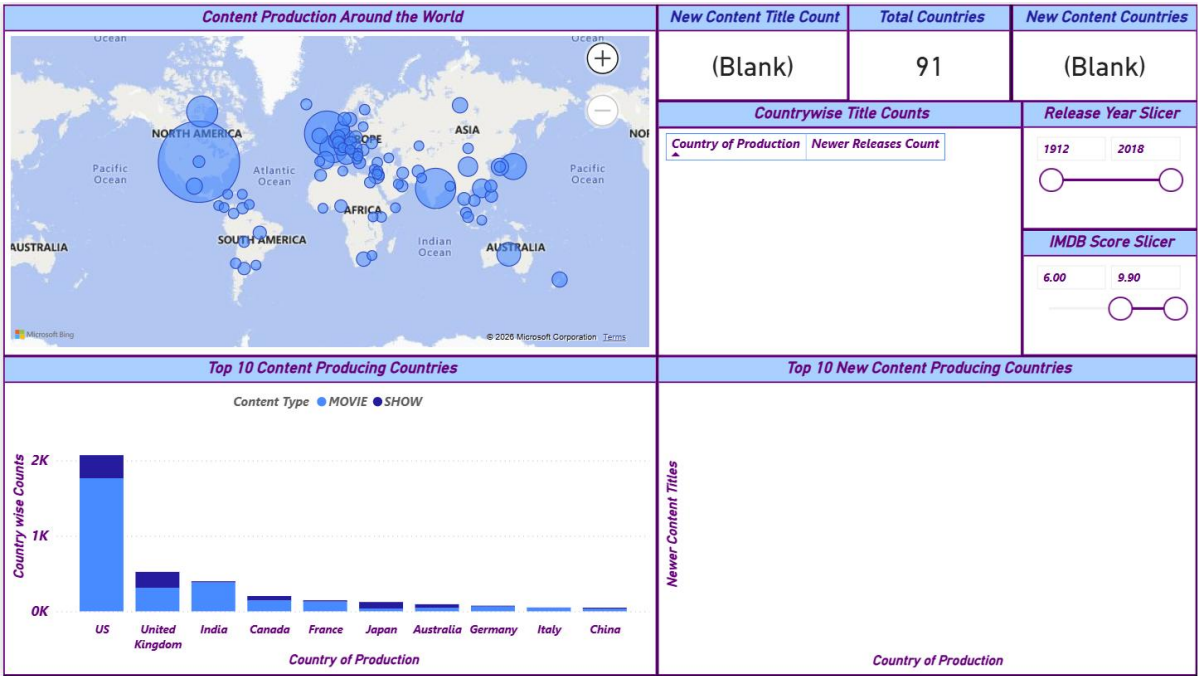
When used slicer to show high rated content i.e. imdb score >= 6.0



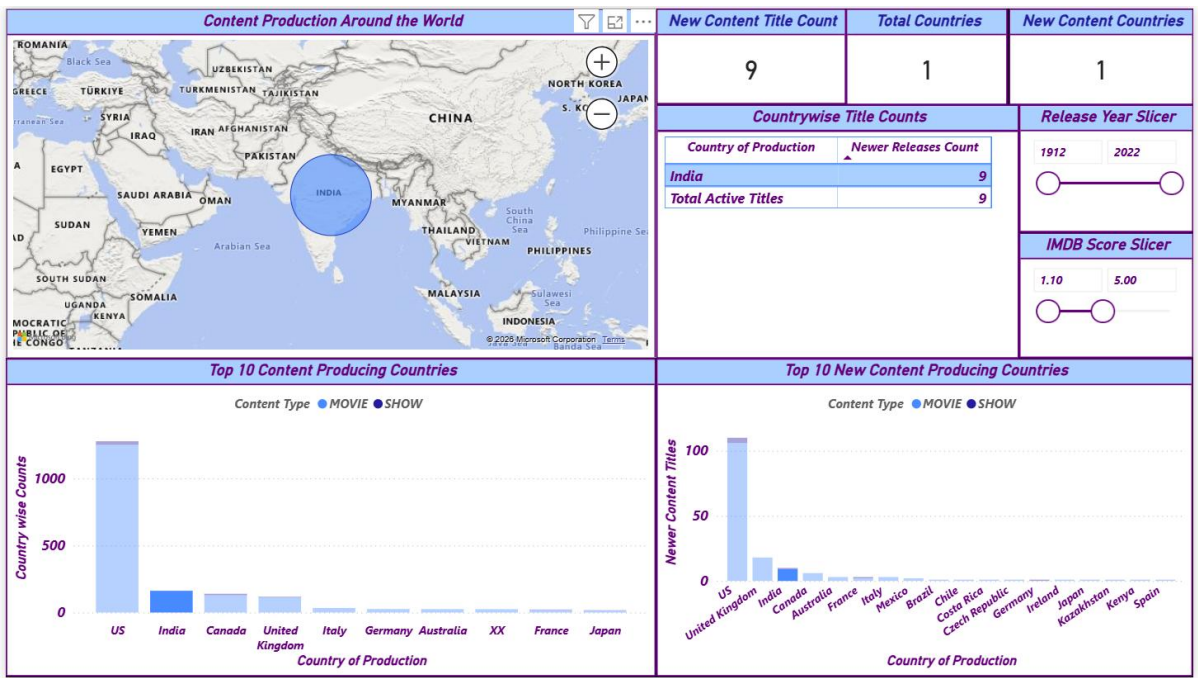
When used slicer to show high rated content i.e. imdb score >= 6.0 and selected countries US, UK, India, Japan, Canada and Australia on the map



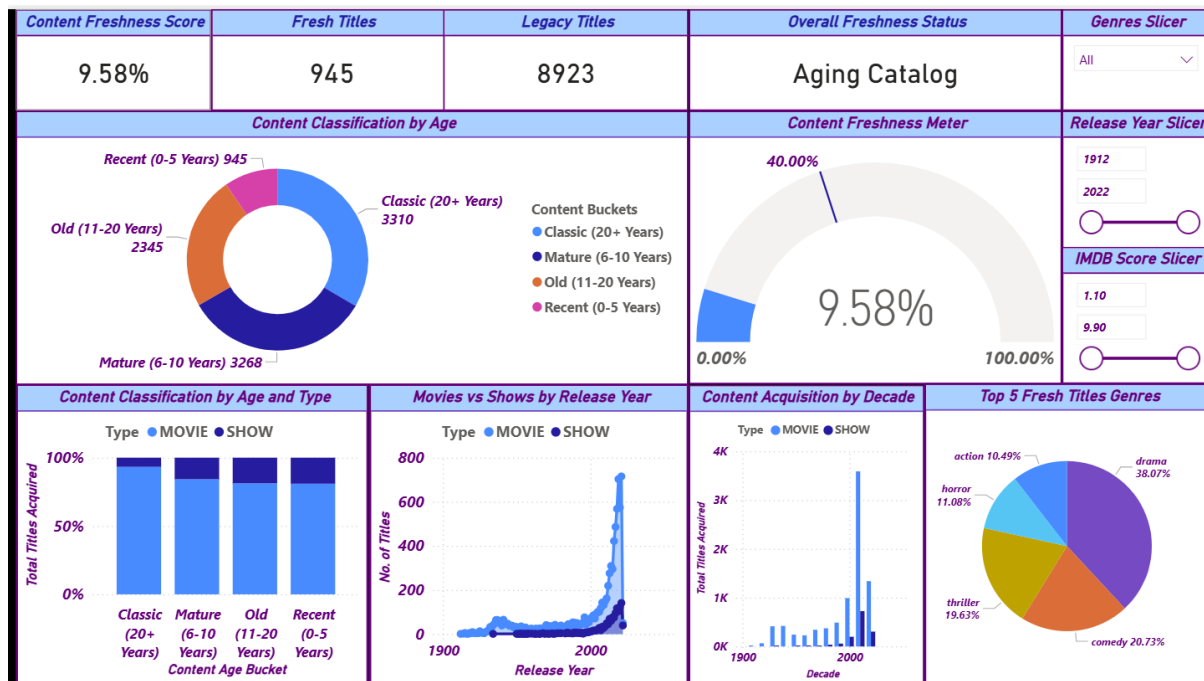
When used slicer to show high rated content i.e. imdb score ≥ 6.0 for legacy titles (i.e. release year ≤ 2018)



When used slicer to show lower rated content i.e. imdb score ≤ 5.0 and with selection of one specific country (India in this case)



Screen 4: Content Freshness & Lifecycle Analysis



Purpose:

Measures content age and evaluates catalog freshness.

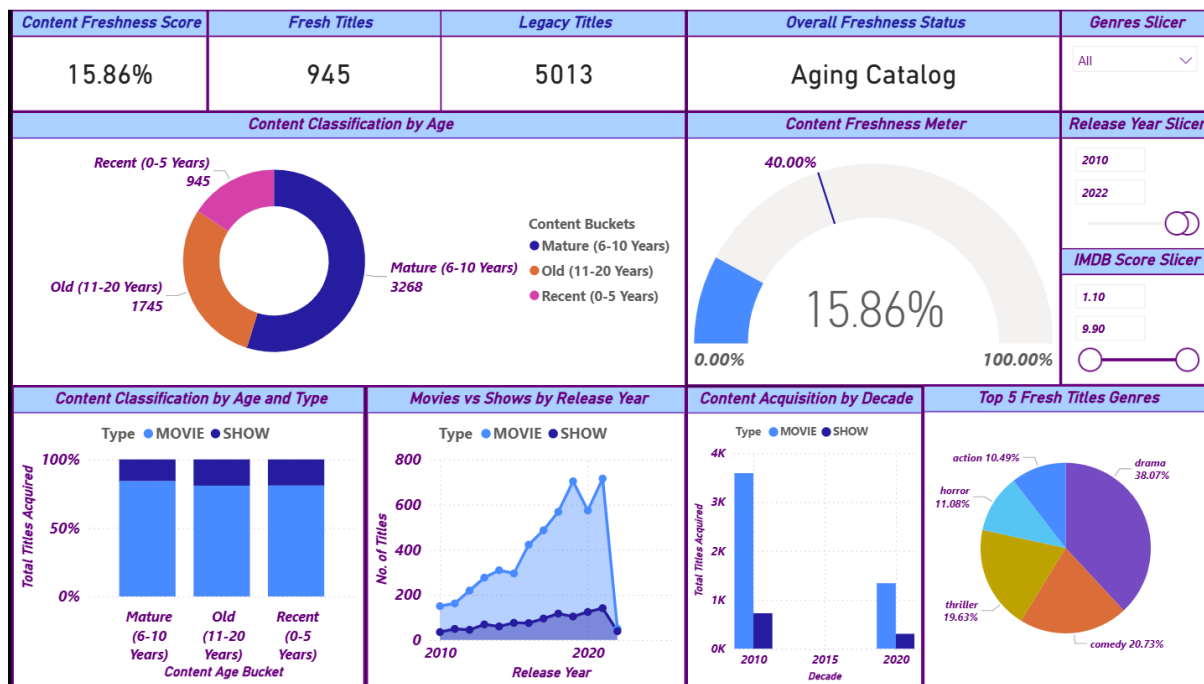
Key Visuals:

- Release Year Distribution
 - Decade Wise Content Acquisition
 - Content Age Segmentation
 - New vs Legacy Content Analysis
-
- **KPI cards** provide an executive summary of the freshness score, recent titles, legacy titles, and overall catalog status.
 - A **donut chart** shows how the library is distributed across different content age groups, while
 - A **gauge** visually indicates whether the freshness level meets business expectations here in this case 40% is the target.
 - **Stacked column chart** to compare the proportion of Movies and TV Shows within each age category and
 - A **line chart** to analyze release trends over time by content type.
 - A **clustered column chart** to identify which decades contributed the most content and finally, a pie chart highlights the genres dominating recent acquisitions.

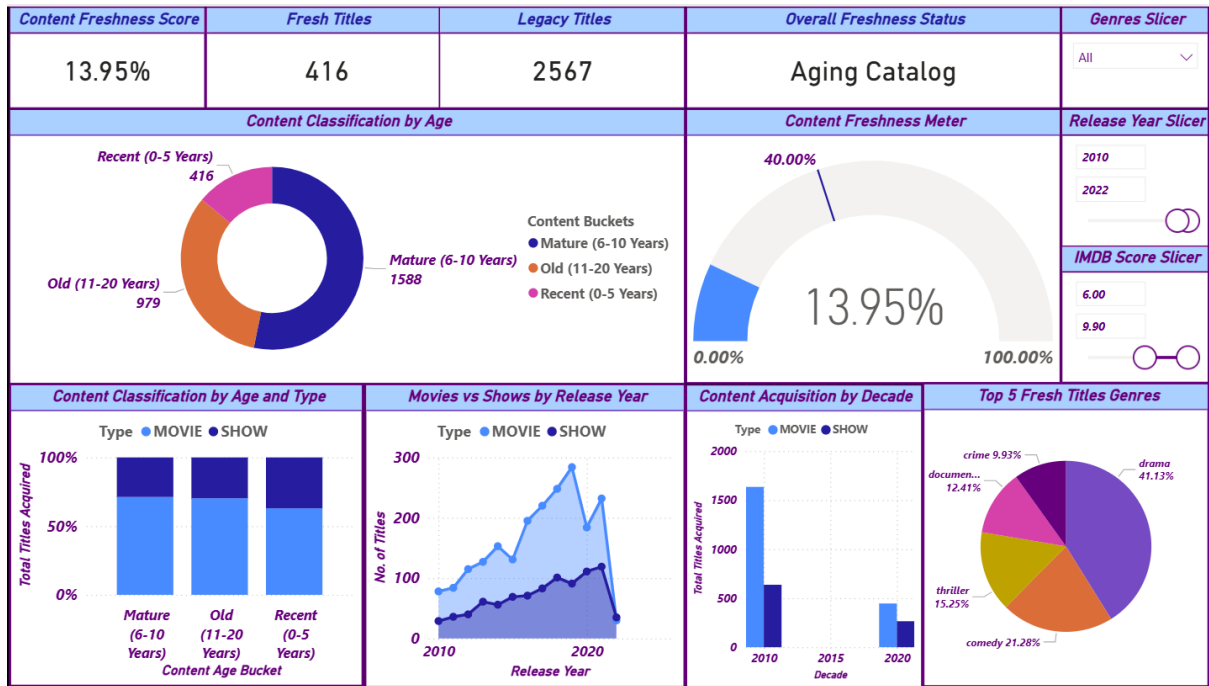
- Together, these visuals help decision-makers assess catalog health, identify aging content, and plan future acquisition strategies to maintain a competitive streaming library

Other Screenshots after various user interactions through slicers:

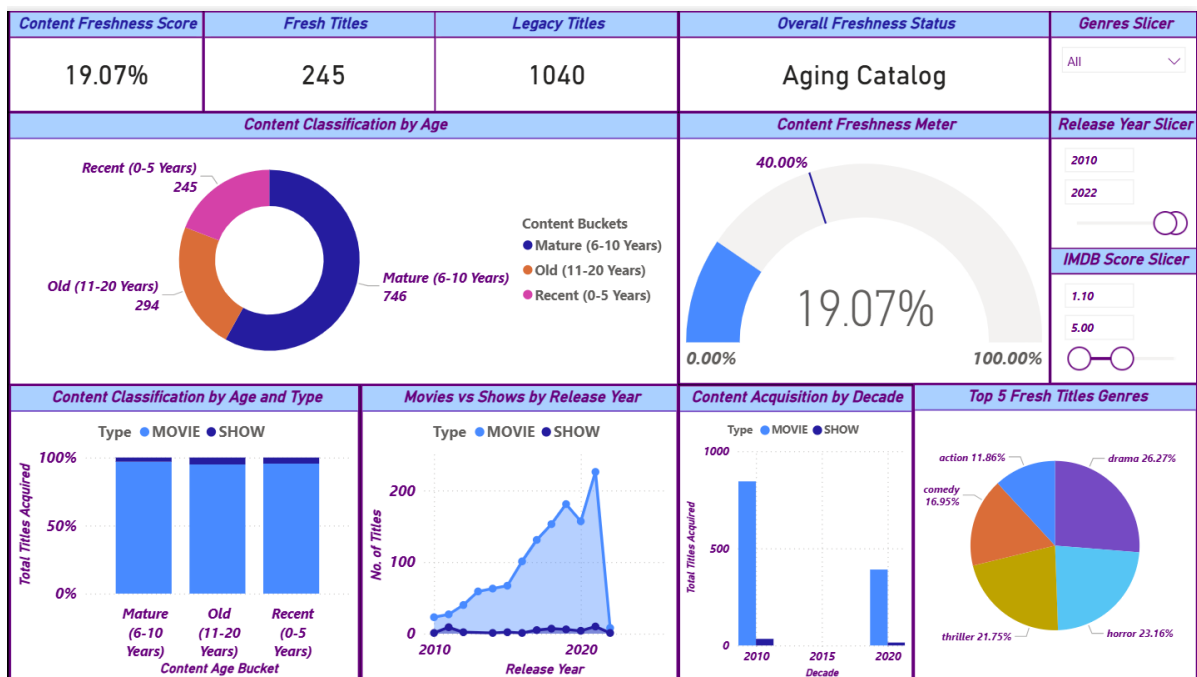
When used slicer to show content for release_year >=2010



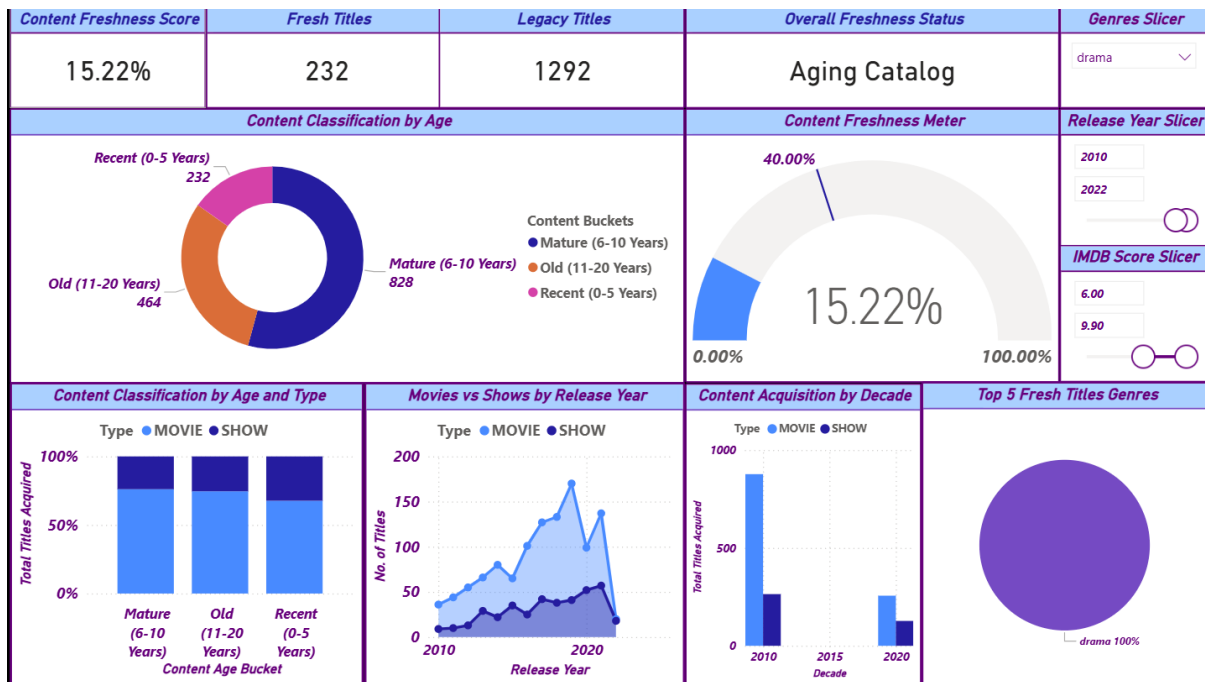
When used slicer to show content for release_year >=2010 and highly rated content imdb score >= 6.0



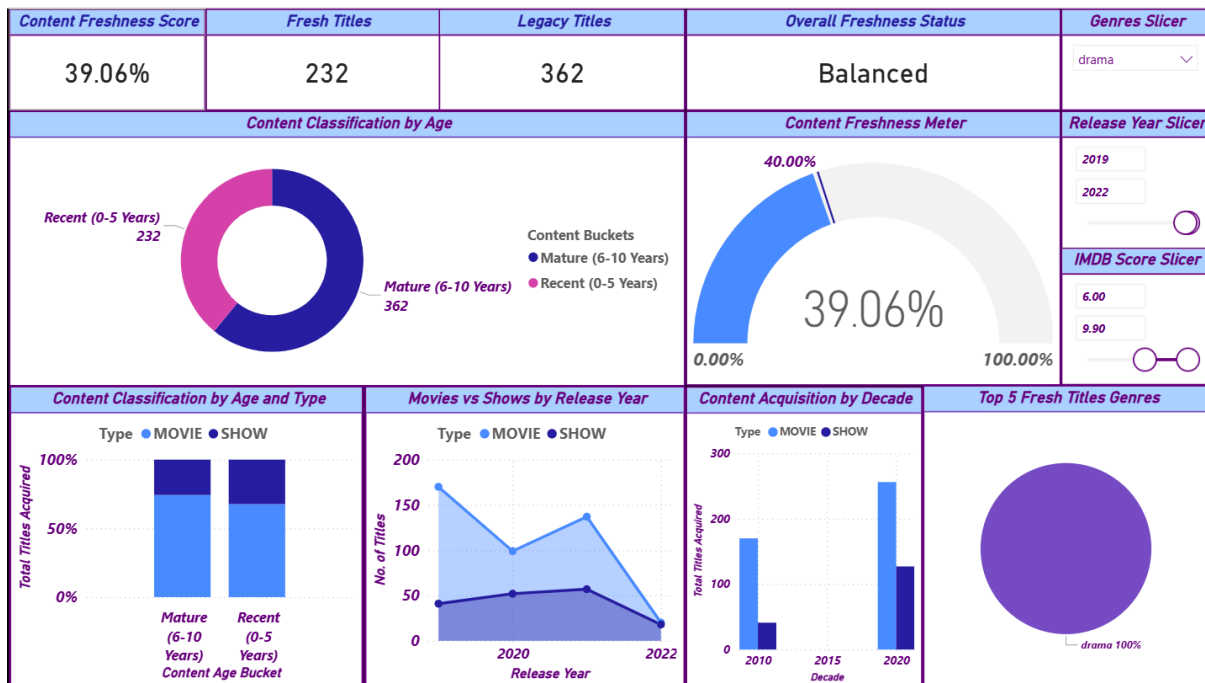
When used slicer to show content for release_year ≥ 2010 and low rated content imdb score < 5.0



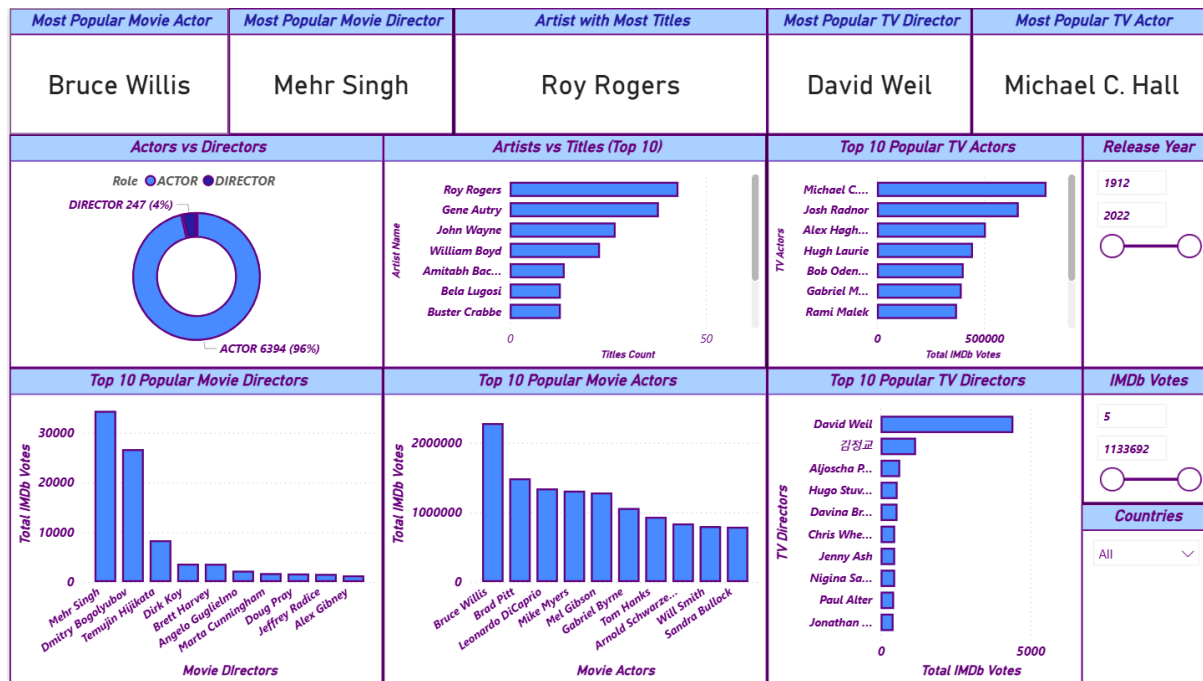
When used slicer to show content for release_year ≥ 2010 and highly rated content imdb score ≥ 6.0 for drama genre



When used slicer to show latest content for release_year >=2019 and highly rated content imdb score >= 6.0 for drama genre



Screen 5: Artist Talent Analysis



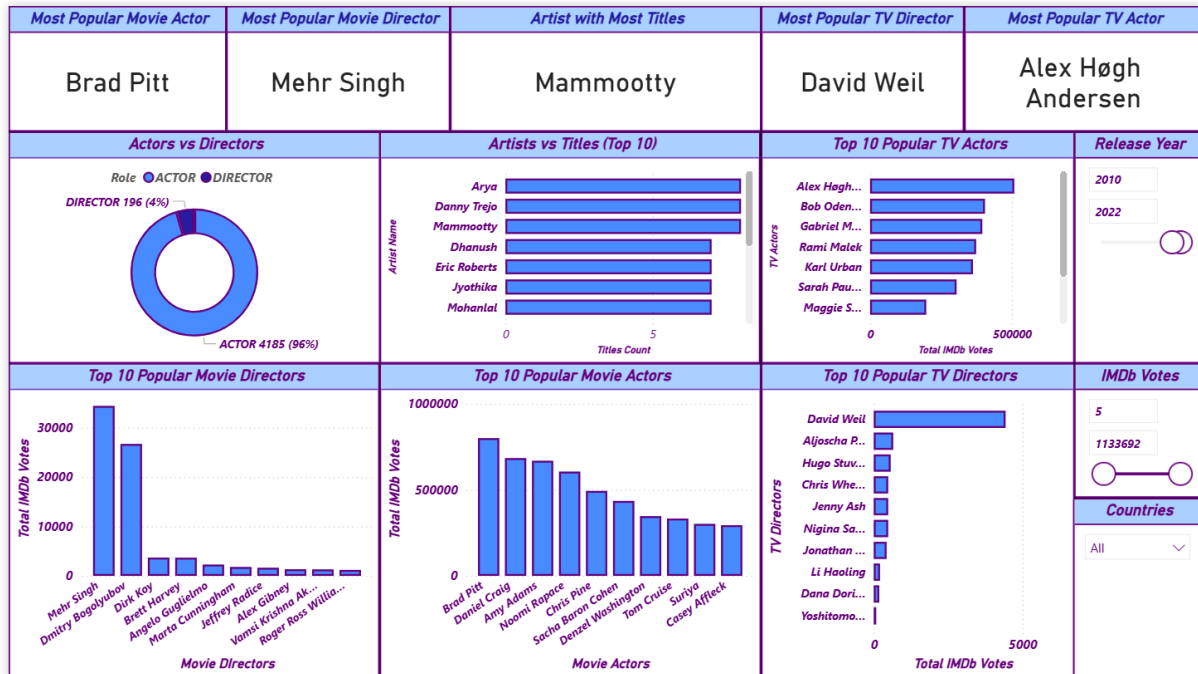
Purpose:
Evaluates audience coverage and genre diversity.

Key Visuals:

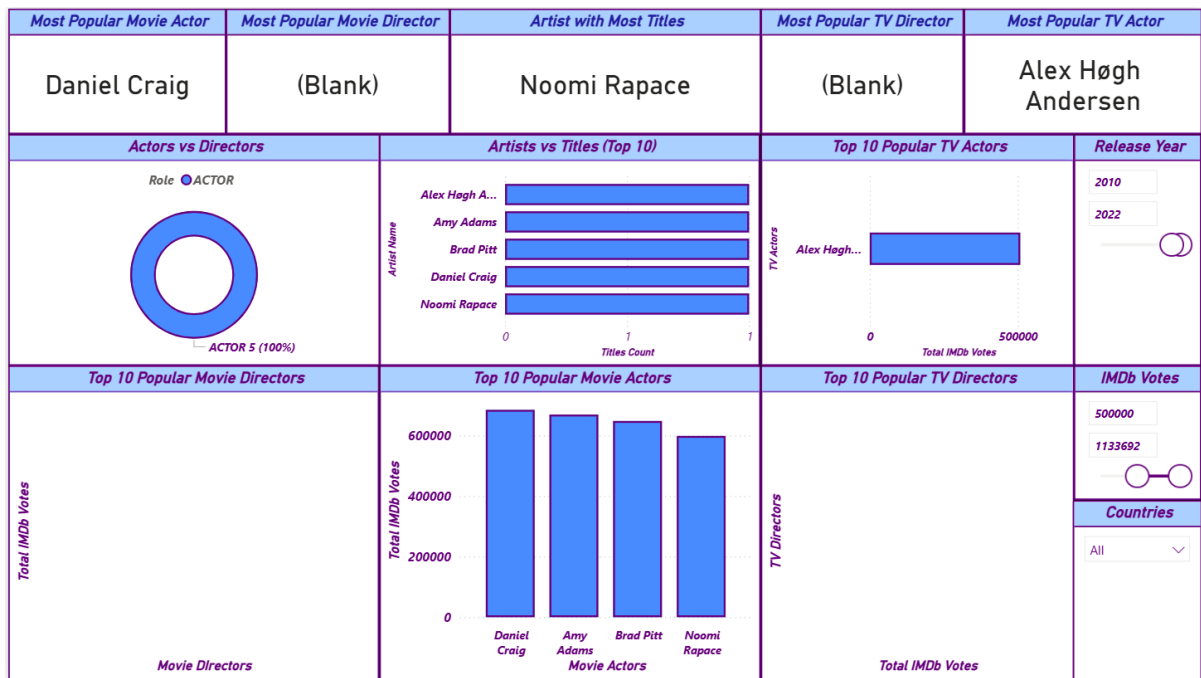
- Top 10 Popular Movie and TV Actors
 - Top 10 Popular Movie and TV Directors
 - Actors vs Directors
 - Top 10 Artists with most no. of titles
-
- This dashboard to evaluate the performance and contribution of actors and directors within the Amazon Prime Video catalog.
 - **KPI cards** provide an executive summary by identifying the most popular movie actor, movie director, TV actor, TV director, and the artist with the most titles.
 - A **donut chart** shows the overall distribution of actors and directors in the dataset.
 - **Horizontal and vertical bar charts** to rank artists by title count and IMDb popularity because they make comparisons across multiple individuals clear and easy to interpret.
 - This along with **sliders** for release year, IMDb votes, and country allow users to explore talent performance across different time periods, popularity levels, and production markets.
 - Overall, these visuals help identify influential talent, support casting and acquisition decisions, and reveal opportunities for strategic partnerships.

Other Screenshots after various user interactions through sliders:

When used slicer to show latest content for release_year >=2010



When used slicer to show latest content for release_year >=2010 and with artist who secured imdb votes >= 500000



When used slicer to show latest content for release_year >=2010 and with artists who content was produced in “Canada”



Final Tab: Conclusion

Insights, Recommendations and Business Impact	
Insights: • Movies dominate the content library. • The average IMDb rating is 6.01, indicating an overall moderately rated catalog. • The average content age is 24.65 years which indicates the content library is primarily outdated. • Drama clearly dominates the catalog followed by Comedy, Thriller, Action etc. • Documentation, Reality, History, War and Animation perform better than the overall average rating of 6.01 • Family-oriented certifications represent a smaller share of the catalog. • North America and Europe contribute the largest share of titles, with contributions from Asia and other regions as well.	
Business Recommendations: • Increase the proportion of TV Shows to improve subscriber retention. • Prioritize acquiring titles with IMDb ratings above 7.0. • Expand the catalog with more recent releases while retaining selected evergreen classics. • Continue investing in Drama while increasing high-rated genres such as Documentaries, History, and Animation. • Promote highly rated but underexposed titles using personalized recommendations. • Expand children's and family programming to diversify the audience base. • Increase localized content acquisitions from emerging international markets.	
Business Impact Summary:	
Recommendation • Invest in high-performing genres • Refresh the content library • Expand original TV shows • Promote hidden high-rated titles • Diversify regional acquisitions • Increase family-friendly programming • Focus on high-quality content	Expected Impact - Higher viewer engagement and better ROI - Improved subscriber acquisition and retention - Increased watch time and reduced churn - Better utilization of existing content - Stronger international market growth - Broader audience reach and more household subscriptions - Sustained user satisfaction and stronger platform reputation

Purpose:

Provides summary of Insights, Strategic recommendations and the associated Business Impact details.

No Visuals but a text box control with these details. Quick Reference for business stakeholders.

5. Key Insights and Explanations

Insight 1: Movies Dominate the Catalog

The majority of content available on Amazon Prime Video consists of movies, indicating a stronger investment in movie acquisitions compared to TV shows.

Insight 2: Genre Concentration

Drama, Comedy, and Documentary categories account for a significant portion of the content library, while niche genres have lower representation.

Insight 3: Geographic Concentration

Content production is concentrated among a few countries, suggesting opportunities to expand content sourcing from underrepresented regions.

Insight 4: Aging Content Portfolio

A substantial percentage of titles were released more than five years ago, highlighting the need for regular content refresh strategies.

Insight 5: Expansion Opportunities

Several countries and genres show relatively low representation, indicating potential areas for future investment and audience growth.

Insight 6: Audience Coverage Gaps

Certain audience segments may be underserved due to limited availability of specific genres and content ratings.

6. Business Impact Summary

This project provides a centralized analytics platform for evaluating Amazon Prime Video's content portfolio and supporting strategic business decisions.

Expected Business Benefits

- Improved content acquisition planning
- Better allocation of content investment budgets
- Identification of underserved markets and genres
- Enhanced global expansion strategies
- Increased content diversity and audience coverage
- Better monitoring of catalog freshness
- Data-driven executive decision-making

Strategic Recommendation

Amazon Prime Video should focus on:

1. Expanding content acquisition in underrepresented regions.
2. Increasing investment in emerging and underserved genres.
3. Maintaining a healthy balance between movies and TV shows.
4. Continuously refreshing the content catalog with newer releases.
5. Leveraging analytics-driven recommendations for future content investments.

Tools & Technologies Used

- Power BI
- SQL

- Python (Pandas, NumPy, Matplotlib)
- Excel

Skills Demonstrated

- Data Cleaning
- Data Modeling
- SQL Analytics
- DAX Calculations
- Data Visualization
- Business Intelligence
- Dashboard Design
- Storytelling with Data
- Strategic Analysis